2. Requirement Analysis

Functional Requirements

- Allow a new user to register with name, email, password and role (farmer, researcher, agribusiness, government).
- Allow a registered user to log in and log out; keep them signed in via a session.
- Let a user download a styled PDF report of a recommendation immediately after it is generated.
- Load and read the crop recommendation dataset.
- Perform preprocessing on soil and environmental data.
- Handle missing values.
- Train and compare multiple Machine Learning models.
- Save the best-performing trained model, scaler and label encoder.
- Accept user inputs (Nitrogen, Phosphorous, Potassium, Temperature, Humidity, Soil pH, Rainfall) through the Smart Recommendation web form.
- Validate that all fields are provided and within their allowed range before predicting.
- Store every submission as a SoilData record linked to the submitting user.
- Recommend the most suitable crop with a confidence percentage, linked to a Crop reference record and the MLModel that produced it.
- Automatically generate a Report record (summary + agronomy recommendations) immediately after each prediction.
- Provide a separate Suitability Assessment tool that scores a chosen crop against given field conditions.
- Provide a Crop Library with details for each supported crop.
- Provide a Research & Policy dashboard with prediction-volume and top-predicted-crop analytics.
- Log every recommendation to a Prediction History page (searchable by crop).
- Export the Prediction History as a CSV file that matches exactly what is currently searched/filtered on screen (not always the full history).

Non-Functional Requirements

- User-friendly interface.
- Secure password storage (hashed, never plain text)
- Fast prediction response.
- Reliable prediction accuracy.
- Clear validation and error messaging.
- Easy maintenance and scalability; normalized data model to avoid redundancy.

Software Requirements

- Python 3.x
- Anaconda Navigator
- Jupyter Notebook / Spyder

- Visual Studio Code / PyCharm
- Flask
- Werkzeug (password hashing)
- NumPy
- Pandas
- SciPy
- Matplotlib
- Seaborn
- Scikit-learn
- Joblib
- ReportLab
- Requests

Hardware Requirements

- Intel Core i3 or above
- Minimum 4 GB RAM
- Minimum 10 GB free storage
- Windows / Linux / macOS
- Internet connection for downloading datasets and packages

Dataset Requirements

- Crop Recommendation Dataset
- 2200 Records, 22 Crop Classes, 8 Attributes
- CSV Format (Crop_recommendation.csv)

Expected Output

Recommendation of the most suitable crop, with a confidence percentage, based on user-provided soil nutrient and environmental indicators, with a Report record generated automatically for every recommendation.